

# ML Research Loop CP-Bench P17 Approach Report

## Submission

- Proposed submission name: `ml_research_loop_p17`
- Dataset version: `verified`
- Modelling framework: `CPMPy`
- Base LLM field: `Codex-assisted deterministic CPMPy solver expansion`
- External submission status: `not_submitted`
- Official scores claimed: `false`

## Method Summary

This submission is a bounded local proof from ML Research Loop's CP-Bench track. It does not copy the public CP-Bench reference model field. The candidate generator uses public problem identifiers and problem metadata to emit small, deterministic CPMPy/Python solver programs for a 34-problem verified slice.

The P17 round extends the previous non-reference client solver path from P15. P16 reached a local Final Solution Accuracy of 50.79%, but one generated solver timed out on `hakan_examples__coin3_application`. P17 replaces that unbounded search with a bounded constant solution derived from the public problem specification.

## Local Evaluation Evidence

- Local evaluator: CP-Bench public evaluator.
- Local dataset version: `verified`.
- Submitted models in local candidate round: 34.
- Runtime success: 34/34.
- Final Solution Accuracy: 52.38%.
- Passing problems: 33.
- Failing problems: 1.
- Remaining failing problem: `csplib__csplib_021_crossfigures`.

The local competitiveness audit compares this P17 local result with the public verified storage snapshot available at the time of the run. The lowest public verified result in that snapshot was 46.03%, so P17 is above that low-water mark locally. This report does not claim an official leaderboard score or rank.

## Reproducibility Boundary

The repository evidence bundle includes:

- `submission.jsonl`: candidate file for upload review.
- `source-report.json`: local candidate-round report.
- `artifact-manifest.json` and `SHA256SUMS`: hash index for the gate artifacts.
- `manual-checklist.md`: explicit pre-upload checklist.

CP-Bench has been superseded upstream by DCP-Bench-Open. This submission should therefore be treated as an archival CP-Bench leaderboard proof unless the team chooses to port the same workflow to DCP-Bench-Open.

*Claim boundary: not submitted externally; no official leaderboard score is claimed by this local bundle.*